

EEE 416 – Microprocessor and Embedded Systems Laboratory
Jan 2026 Level-4 Term-I Section A2
Project Proposal

AI Driven Bangla Braille Learning Device

Submitted by – Group 08

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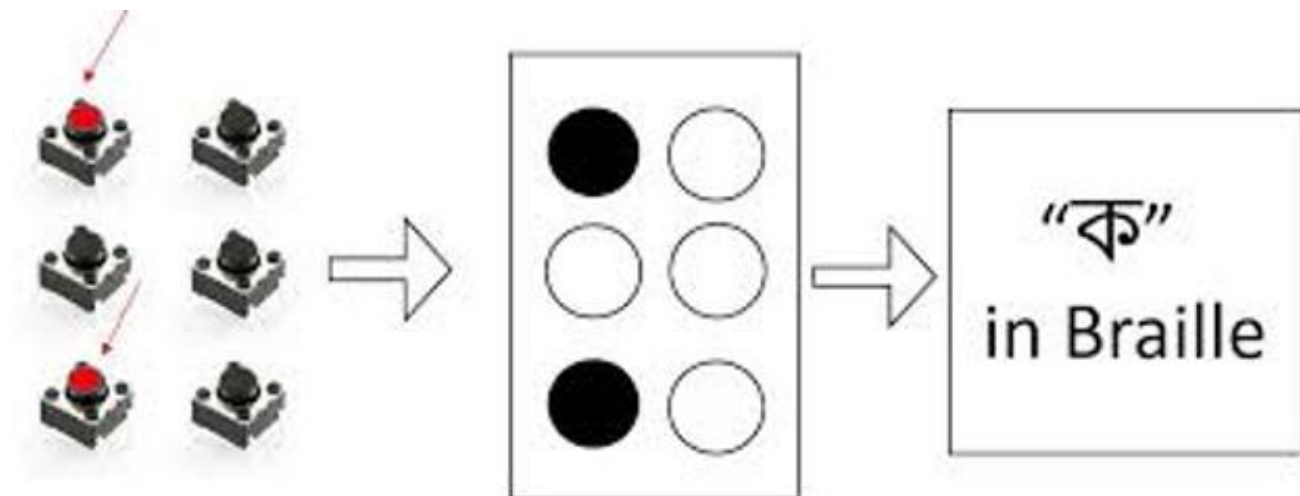
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2.Objectives

- Build an interactive device for learning Bangla Braille.
- Let users learn Bangla Braille by using vibration and audio feedback.
- Let users practice Braille using push buttons.
- Use AI to adapt learning based on user performance.
- Keep the system low-cost and user friendly.



3.Literature Review

Hardware-Based Braille Learning Systems

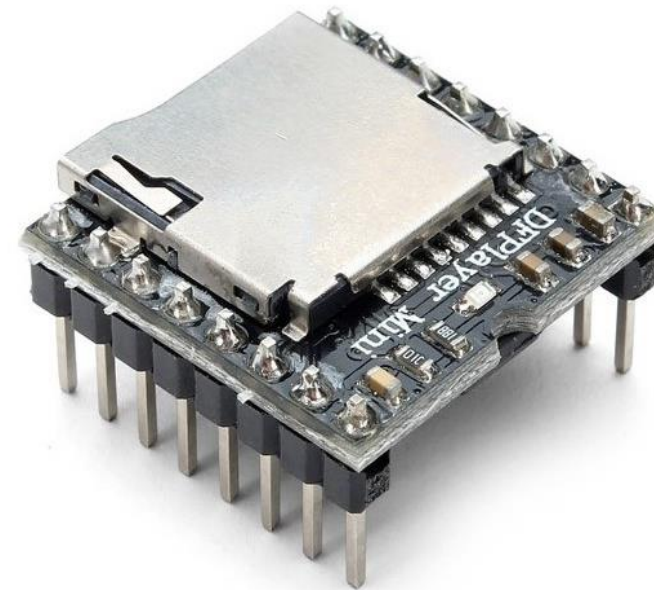
Paper	Main Work	Research Gap for Our Project
Kader et al. (2018) Developing A Self-Learning Braille Kit for Visually Impaired People	Built a self learning Bengali Braille kit using six actuators and audio. It supports learning mode and practice mode.	Costly and it does not include ML/AI based implementation.
Saikot & Sanim (2022) Refreshable Braille Display With Adjustable Cell Size for Learners With Different Tactile Sensitivity	Developed a singlecell refreshable Braille display with six electromechanical actuators, adjustable dot spacing, and speech support.	Expensive.It does not have practice mode.It doesn't include ML/AI based implementation as well.
Ma, Lai & Luo (2025) BrailleRhythm: Enhancing Braille Character Recognition and Learning via Real-time Tactile Sensing and Audio Feedback	Placed a pressure-sensor sheet over Braille cards. The system detects touched Braille patterns and gives real-time audio feedback.	It is not Bangla focused and does not test learner answers, retries, or adaptive teaching actions.

4.Methodology

Hardware Architecture



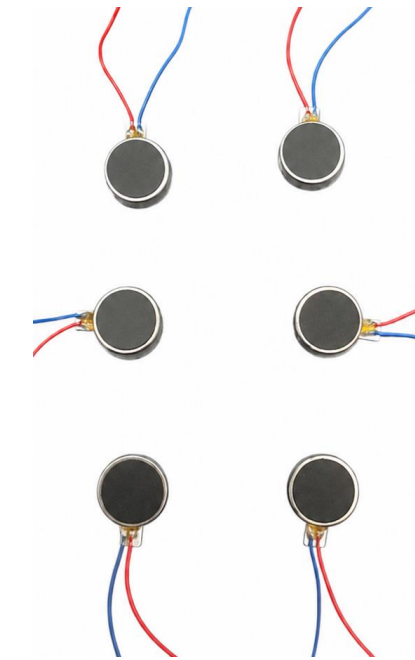
**ESP32 Dev Kit
(MCU)**



**DFPlayer Mini
(Audio Feedback)**



**Tactile Push Buttons
(User Braille Input)**



**1030 Coin Vibration Motor
(Haptic Feedback)**

Methodology

Haptic Teaching Module

Haptic Teaching Module:

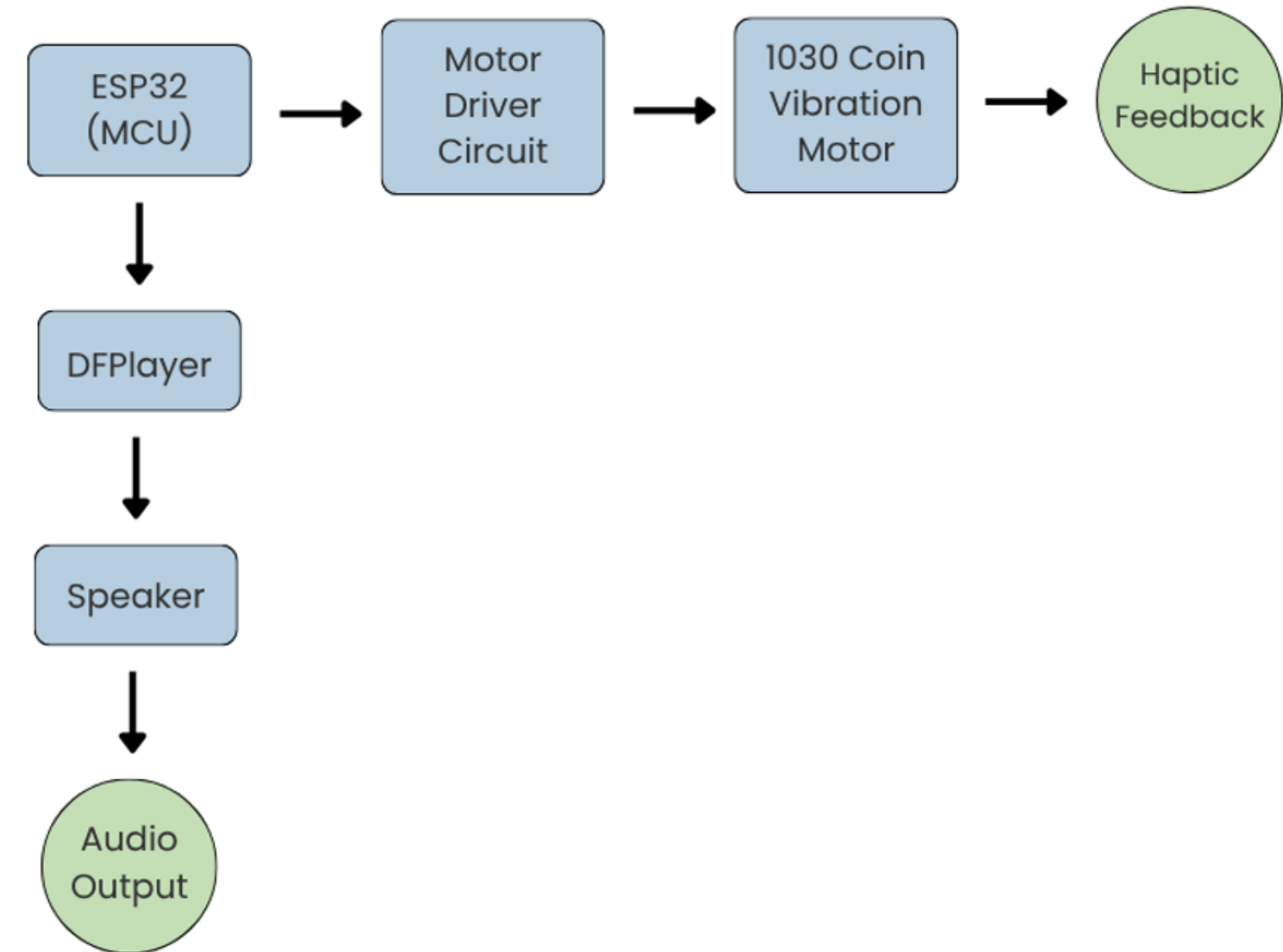
This module teaches Bangla Braille using both audio and vibration, so the learner can hear the letter and feel its Braille pattern.

Auditory Pathway:

The ESP32 selects a letter and sends a command to the DFPlayer, which plays the corresponding Bangla audio through the speaker.

Haptic Pathway:

At the same time, the ESP32 controls the motor driver to activate the required combination of six vibration motors, creating the Braille pattern as haptic feedback.



Methodology

Interactive Evaluation Module

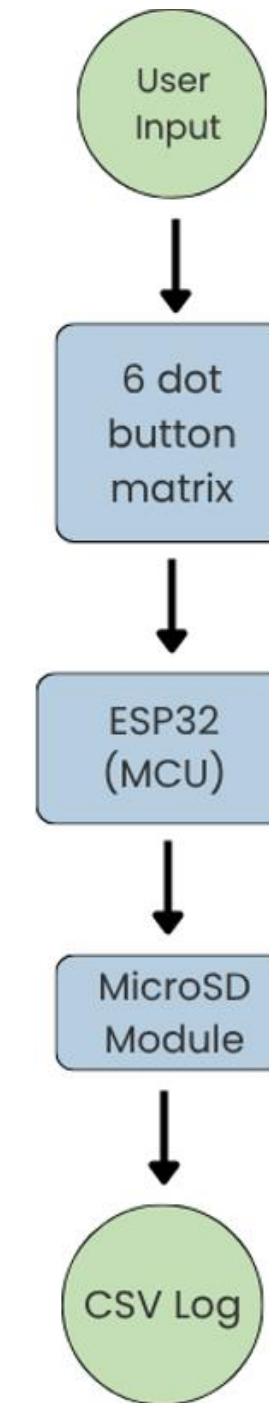
Real-Time Input & Assessment

i. Tactile Input Pathway:

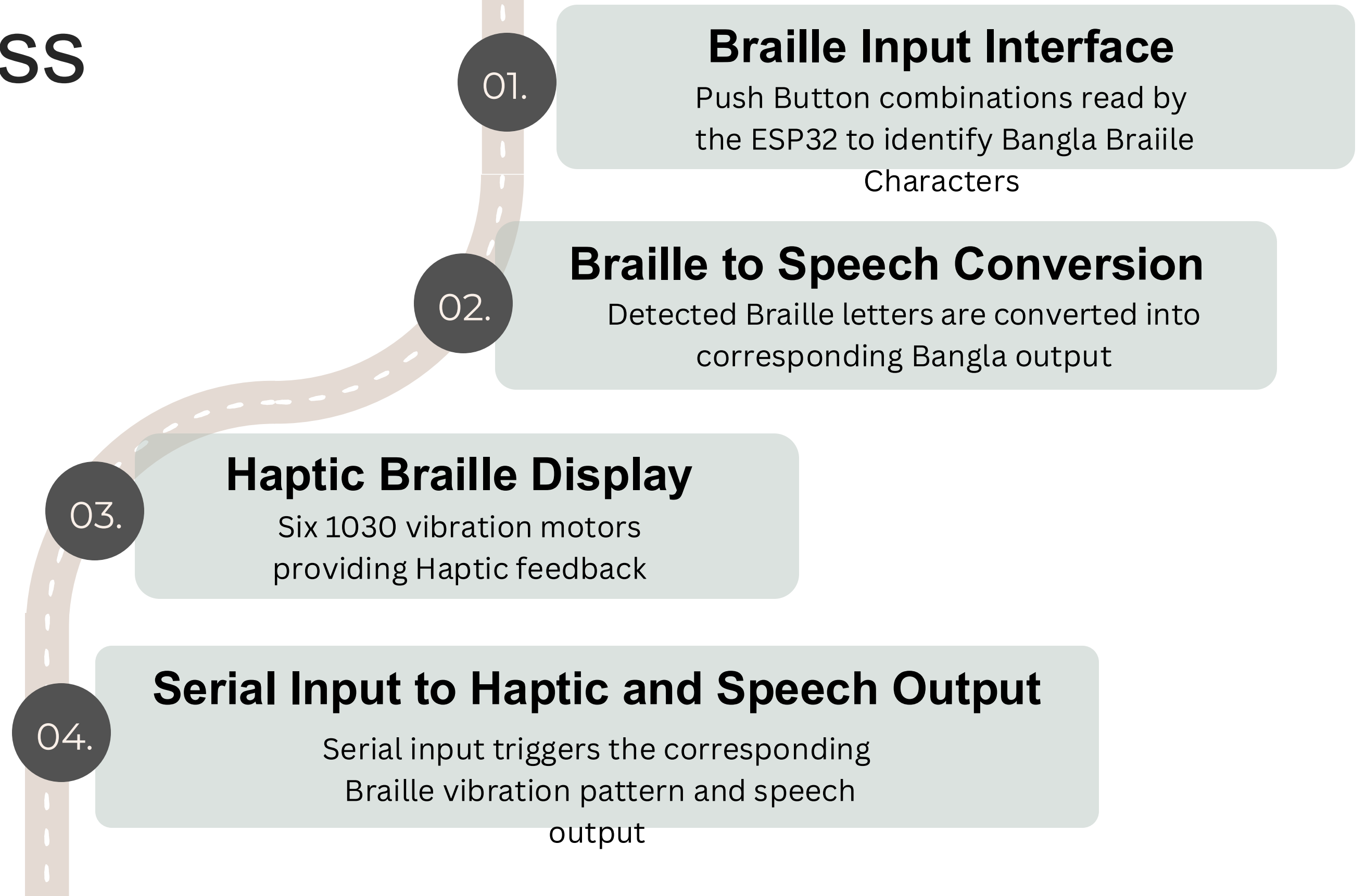
The learner enters a Braille character using the 6 dot button matrix. The ESP32 reads the input and records features.

ii. Data Processing & Logging:

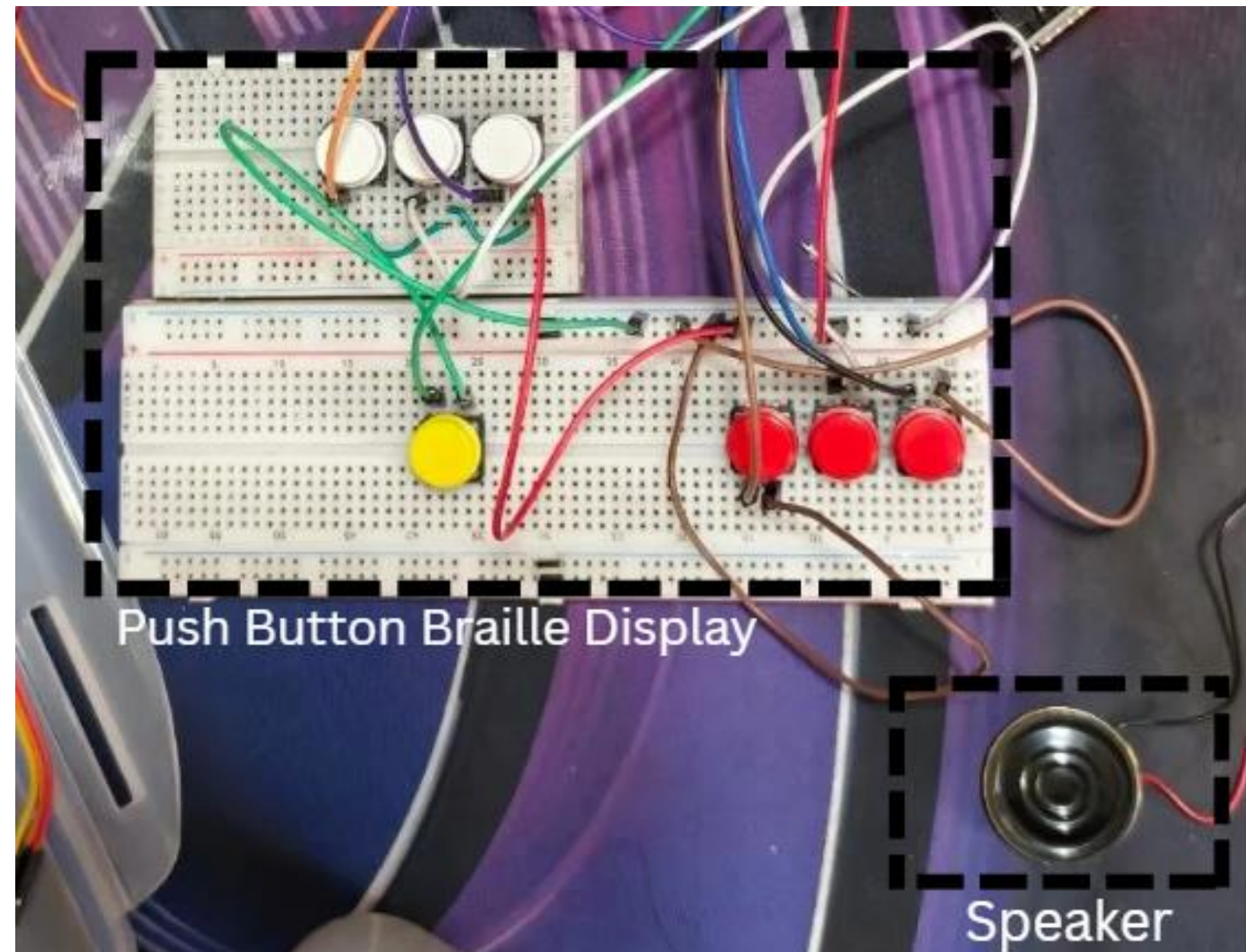
The ESP32 processes the 6-bit Braille pattern and stores the attempt result and collected features in a CSV file through the microSD module for further evaluation.



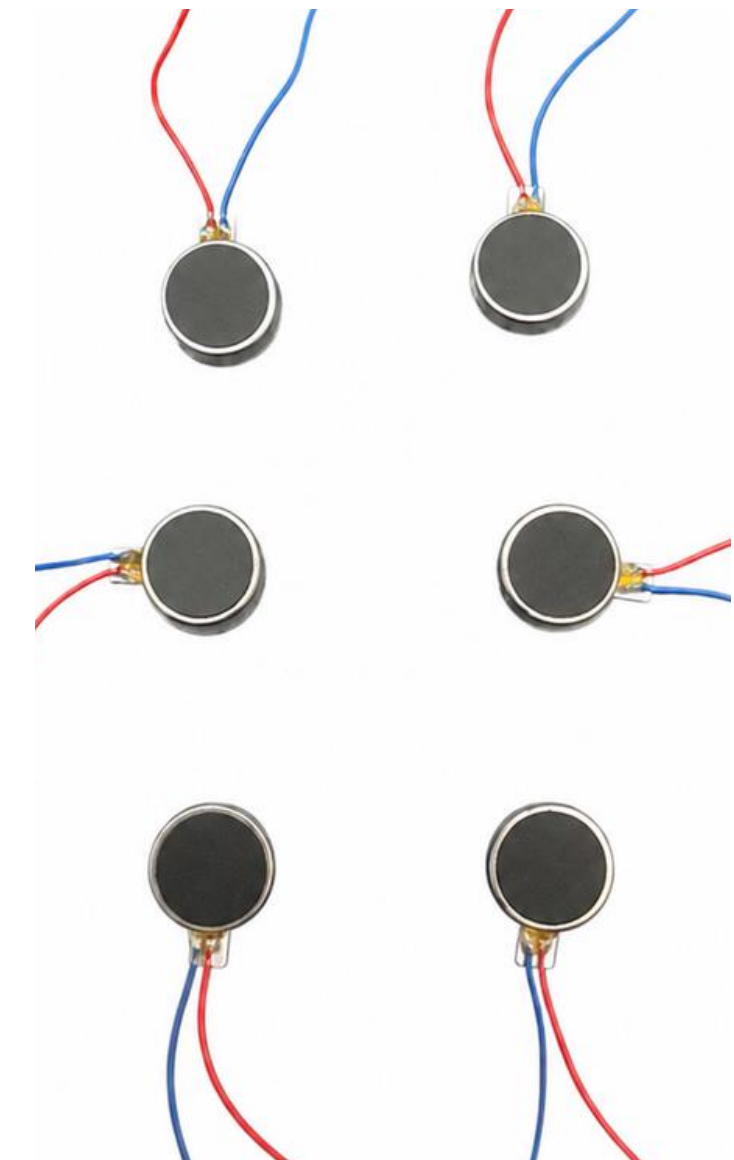
5 Progress



5 Progress



Push Button Braille Display with Speaker



**1030 Coin Vibration Motor
(Haptic Feedback)**

5 Progress

বাংলা ব্রেইল Bangla Braille Tutor — AI Data Collection synced · 77 rows

● SESSION SETUP

Participant code: P03 Collection mode: Normal practice Attempts this session: 20 Character set: All letters (50)

Start session Export CSV Retry queued rows Reset this participant

Weighted toward weaker characters. Produces the natural class distribution.

● ATTEMPT 1 / 20 · DIFFICULTY 1

ট

tta — enter the dots you hear

placeholder pattern

1 4 2 5 3 6

Dot 1 F / Num 7 Dot 4 J / Num 8 Dot 2 D / Num 4 Dot 5 K / Num 5 Dot 3 S / Num 1 Dot 6 L / Num 2

Submit Clear esc Hint H End session

Replay

● PER-CHARACTER MASTERY

CHAR	NAME	SEEN	ACC	MASTERY	STREAK	MISTAKES
ঁ	chandrabinu	6	17%		-3	5
ধ	dha	2	50%		+1	1
ড়	rra	5	20%		+1	4
তা	aa	4	50%		+2	2
ট	tta	4	50%		+2	2

● SESSION SETUP

Participant code: P03 Collection mode: Normal practice Attempts this session: 20 Character set: All letters (50)

Start session Export CSV Reset this participant

Weighted toward weaker characters. Produces the natural class distribution.

Normal practice Targeted — drive rare classes Review — weak characters only

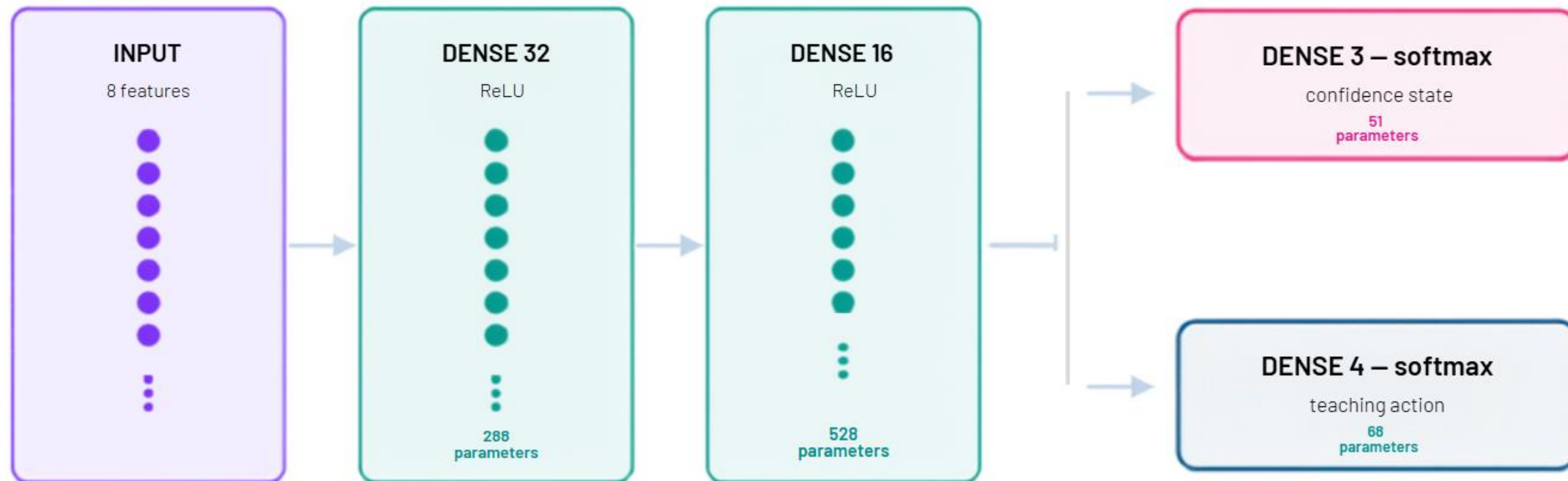
Web Platform for Practice and ML Data Collection



Remaining Task –1 : Testing Phase

- 1.Letter Sound:** The speaker plays the sound of a target letter.
- 2.User Response:** The user listens to the sound and presses the corresponding push button.
- 3.System Evaluation:** The system checks the selected button against the correct letter.
- 4.Feedback:** The result is provided through **vibration and/or speaker feedback**.
- 5.Learning Update:** The response is recorded for future learning and performance analysis.

Remaining Task –2 : Model Architecture



Remaining Task –2 : Model Architecture

Previous proposed Inputs (14)
char_id
response_time
press_duration
retry_count
prev_accuracy
prev_mastery
hint_count
session_number
difficulty_level
time_since_last_practice
prev_confidence
current_streak
wrong_streak
prev_mistakes



Updated Inputs (8)
response_time
press_duration
retry_count
prev_accuracy
prev_mastery
hint_count
current_streak
wrong_streak

Remaining Task –2 : Model Architecture

Previously proposed Output
(teaching_action)

REPEAT

HINT

NORMAL_PRACTICE

INCREASE_DIFFICULTY

REVIEW_PREVIOUS

WORD_PRACTICE



Updated Output
(teaching_action)

REPEAT

HINT

NORMAL_PRACTICE

REVIEW_PREVIOUS

Remaining Task –3 : 3d model design



[N.B.: This is an AI-generated concept design for visualization purposes only]